SOLVE AND CHECK

Name: ..... **Date:** .....

Solve each equation in two steps: undo the number added or subtracted, then divide. Then CHECK by putting your answer back into the original equation and showing both sides match.

1. **3x − 11 = -8**

Step 1 _____

Step 2 _____

Check _____

2. **9x + 7 = 16**

Step 1 _____

Step 2 _____

Check _____

3. **3x + 6 = 24**

Step 1 _____

Step 2 _____

Check _____

4. **2x + 3 = 15**

Step 1 _____

Step 2 _____

Check _____

5. **9x + 1 = 55**

Step 1 _____

Step 2 _____

Check _____

6. **2x − 13 = -3**

Step 1 _____

Step 2 _____

Check _____

7. **6x + 5 = 23**

Step 1 _____

Step 2 _____

Check _____

8. **7x + 10 = 94**

Step 1 _____

Step 2 _____

Check _____

9. **6x + 3 = 75**

Step 1 _____

Step 2 _____

Check _____

10. **8x − 2 = 46**

Step 1 _____

Step 2 _____

Check _____

SOLVE AND CHECK ANSWER KEY

Name: ..... Date: .....

Answer key: both steps and the substitution check for every equation.

1. **3x − 11 = -8**

Step 1 3x = -8 + 11 = 3

Step 2 x = 3 ÷ 3 = 1

Check 3(1) − 11 = 3 − 11 = -8 ✓

2. **9x + 7 = 16**

Step 1 9x = 16 − 7 = 9

Step 2 x = 9 ÷ 9 = 1

Check 9(1) + 7 = 9 + 7 = 16 ✓

3. **3x + 6 = 24**

Step 1 3x = 24 − 6 = 18

Step 2 x = 18 ÷ 3 = 6

Check 3(6) + 6 = 18 + 6 = 24 ✓

4. **2x + 3 = 15**

Step 1 2x = 15 − 3 = 12

Step 2 x = 12 ÷ 2 = 6

Check 2(6) + 3 = 12 + 3 = 15 ✓

5. **9x + 1 = 55**

Step 1 9x = 55 − 1 = 54

Step 2 x = 54 ÷ 9 = 6

Check 9(6) + 1 = 54 + 1 = 55 ✓

6. **2x − 13 = -3**

Step 1 2x = -3 + 13 = 10

Step 2 x = 10 ÷ 2 = 5

Check 2(5) − 13 = 10 − 13 = -3 ✓

7. **6x + 5 = 23**

Step 1 6x = 23 − 5 = 18

Step 2 x = 18 ÷ 6 = 3

Check 6(3) + 5 = 18 + 5 = 23 ✓

8. **7x + 10 = 94**

Step 1 7x = 94 − 10 = 84

Step 2 x = 84 ÷ 7 = 12

Check 7(12) + 10 = 84 + 10 = 94 ✓

9. **6x + 3 = 75**

Step 1 6x = 75 − 3 = 72

Step 2 x = 72 ÷ 6 = 12

Check 6(12) + 3 = 72 + 3 = 75 ✓

10. **8x − 2 = 46**

Step 1 8x = 46 + 2 = 48

Step 2 x = 48 ÷ 8 = 6

Check 8(6) − 2 = 48 − 2 = 46 ✓

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